1)

$T(n) = 5T(n/7) + 3n^2$

$T(n/7) = 5T(n/7^2) + 3(n/7)^2$

$T(n) = 5[5T(n/7^2) + 3(n/7)^2] + 3n^2$

$= 25T(n/7^2) + 15(n^2/7^2) + 3n^2$

$= 25T(n/49) + 3n^2(5/49 + 1)$

$T(n/49) = 5T(n/7^3) + 3(n/49)^2$

$= 25[5T(n/343) + 3(n/49)^2] + 3n^2(5/49 + 1)$

$125T(n/343) + 75(n/49)^2 + 3n^2(5/49 + 1)$

After the ath sub for $a = 1$

$T(n) = 5^a T(n/7^a) + 3n^2 \sum_{i=0}^{a-1} (5/49)^i$

Master Theorem

$T(n) = 5T(n/7) + 3n^2$

$a = 5,\ b = 7,\ f(n) = 3n^2,\ d = 2$

$5 < 7^2$ so $\boxed{T(n) = \Theta(n^2)}$

1)

2a) $T(n) = 5T(n/9) + n^7$

$a = 5,\ b = 9,\ f(n) = n^7,\ d = 7$

$5 < 9^7$ so $\boxed{T(n) = \Theta(n^7)}$

2.b) $T(n) = 3T(n/3) + 7n$

$a = 3,\ b = 3,\ f(n) = 7n,\ d = 1$

$3 = 3^1$ so $\boxed{T(n) = \Theta(n \log n)}$

2c) $T(n) = 15T(n/4) + 100$

$a = 15,\ b = 4,\ f(n) = 100(n^0),\ d = 0$

$15 > 4^0$ so $\boxed{T(n) = \Theta(n^{\log_4 15})}$

2d) $T(n) = 9T(n/3) + n^2$

$a = 9,\ b = 3,\ f(n) = n^2,\ d = 2$

$9 = 3^2$ so $\boxed{T(n) = \Theta(n^2 \log n)}$

2e) $T(n) = 6T(n/5) + n^3$

$a = 6,\ b = 5,\ f(n) = n^3,\ d = 3$

$6 < 5^3$ so $\boxed{T(n) = \Theta(n^3)}$

2)

3)

Initial:

BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, LA_GUA, TOL, META, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR

Iteration 1:

Null	BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, TOL, META, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR
A	LA_GUA
B	
C	
D	
E	
F	
G	
H	
I	
J	
K	
L	
M	
N	
O	
P	
Q	
R	
S	
T	
U	
V	
W	
X	
Y	
Z	
Non-alpha	

BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, TOL, META, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR, LA_GUA

Iteration 2:

Null	BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, TOL, META, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	
K	
L	
M	
N	
O	
P	
Q	
R	
S	
T	
U	LA_GUA
V	
W	
X	
Y	
Z	
Non-alpha	

BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, TOL, META, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR, LA_GUA

Iteration 3:

Null	BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, TOL, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR
A	META
B	
C	
D	
E	
F	
G	LA_GUA
H	
I	
J	
K	
L	
M	
N	
O	
P	
Q	
R	
S	
T	
U	
V	
W	
X	
Y	
Z	
Non-alpha	

BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, TOL, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR, META, LA_GUA

Iteration 4:

Null	
A	GUA, AMA, NSA, GUA
B	
C	SUC, VIC,
D	
E	
F	
G	MAG
H	
I	QUI, HUI,
J	
K	
L	ATL, VAL, CAL, TOL, BOL,
M	
N	SAN, SAN
O	CHO
P	
Q	CAQ
R	COR
S	RIS, CES, CAS
T	PUT, ANT, META
U	VAU, CAU,
V	
W	
X	
Y	BOY
Z	
Non-alpha	LA_GUA

GUA, AMA, NSA, GUA, SUC, VIC, MAG, QUI, HUI, ATL, VAL, CAL, TOL, BOL, SAN, SAN, CHO, CAQ, COR, RIS, CES, CAS, PUT, ANT, META, VAU, CAU, BOY, LA_GUA

Iteration 5:

Null	
A	MAG, VAL, CAL, SAN, SAN, CAQ, CAS, VAU, CAU, LA_GUA
B	
C	
D	
E	CES, META
F	
G	
H	CHO
I	VIC, RIS
J	
K	
L	
M	AMA
N	ANT
O	TOL, BOL, COR, BOY
P	
Q	
R	
S	NSA,
T	ATL
U	GUA, GUA, SUC, QUI, HUI, PUT
V	
W	
X	
Y	
Z	
Non-alpha	

MAG, VAL, CAL, SAN, SAN, CAQ, CAS, VAU, CAU, LA_GUA, CES, META, CHO, VIC, RIS, AMA, ANT, TOL, BOL, COR, BOY, NSA, ATL, GUA, GUA, SUC, QUI, HUI, PUT

Iteration 6:

Null	
A	AMA, ANT, ATL
B	BOL, BOY,
C	CAL, CAQ, CAS, CAU, CES, CHO, COR
D	
E	
F	
G	GUA, GUA
H	HUI
I	
J	
K	
L	LA_GUA,
M	MAG, META
N	NSA
O	
P	PUT
Q	QUI
R	RIS
S	SAN, SAN, SUC
T	TOL
U	
V	VAL, VAU, VIC
W	
X	
Y	
Z	
Non-alpha	

Final:

AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, LA_GUA, MAG, META, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC

4)

[62, 62, 69, 70, 4, 103, 85, 27, 86, 40, 96, 77, 23]

62 – initial 5, 0 collisions

0	1	2	3	4	5	6
					62	

62 – initial 5, 1 collision, 1 resize, step 5 to 10

0	1	2	3	4	5	6	7	8	9	10	11	12
				62						62		

13	14	15	16

69 – initial 6, 0 collisions

0	1	2	3	4	5	6	7	8	9	10	11	12
				62		69				62		

13	14	15	16

70 – initial 16, 0 collisions

0	1	2	3	4	5	6	7	8	9	10	11	12
				62		69				62		

13	14	15	16
			70

4 – initial 9, 0 collisions

0	1	2	3	4	5	6	7	8	9	10	11	12
				62		69			4	62		

13	14	15	16
			70

103 – initial 13, 0 collisions

0	1	2	3	4	5	6	7	8	9	10	11	12
				62		69			4	62		
13			14			15			16			
103									70			

85 – initial 3, 0 collisions

0	1	2	3	4	5	6	7	8	9	10	11	12
			85	62		69			4	62		
13			14			15			16			
103									70			

27 – initial 7, 0 collisions

0	1	2	3	4	5	6	7	8	9	10	11	12
			85	62		69	27		4	62		
13			14			15			16			
103									70			

86 – initial 16, 1 collision, step 16 to 1

0	1	2	3	4	5	6	7	8	9	10	11	12
	86		85	62		69	27		4	62		
13			14			15			16			
103									70			

40 – initial 13, 2 collisions, step 13 to 9 to 5

0	1	2	3	4	5	6	7	8	9	10	11	12
	86		85	62	40	69	27		4	62		

13	14	15	16
103			70

96 – initial 14, 0 collisions

0	1	2	3	4	5	6	7	8	9	10	11	12
	86		85	62	40	69	27		4	62		

13	14	15	16
103	96		70

77 – initial 9, 1 collision, step 9 to 8

0	1	2	3	4	5	6	7	8	9	10	11	12
	86		85	62	40	69	27	77	4	62		

13	14	15	16
103	96		70

23 – initial 5, 5 collisions, sequence 5 to 1 to 14 to 10 to 6 to 2

0	1	2	3	4	5	6	7	8	9	10	11	12
	86	23	85	62	40	69	27	77	4	62		

13	14	15	16
103	96		70

7)

4: The time complexity O(n), where n is the number of keys inserted because each insertion runs once to put the input inside the hash map. But Each insertion have O(1) average time.

The space complexity is O(M), where M is the current size of the hash table. Since M grows proportional to n after resizing, this can simplify to O(n).

5: The time complexity is $O(n \times m)$, where n is the number of strings in the array and m is the length of the longest string. This is because we run m passes and in each pass we loop through all n strings to place them into buckets.

Space complexity: O(n), because the 256 buckets collectively hold all n strings at any given pass. The number of buckets is always 256 regardless of input so the only space that grows with input is the storage in each bucket for the n strings per pass.